

Md Farhan Rusafi Alam

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| farhan-alam-biotech-portfolio.vercel.app

| github.com/FarhanAlam2001

PROFESSIONAL SUMMARY

M.Sc. Biotechnology graduate with hands-on research experience in microbiology, molecular biology, and computational biology. Completed independent research on endophyte isolation from *Curcuma amada* and gained practical exposure to toxicology, pharmaceutical biotechnology, stem cell research, and molecular docking through certified internships and courses. Proficient in laboratory techniques including microbial culture, DNA/RNA extraction, gel electrophoresis, and bioinformatics tools. Seeking an entry-level position in biotechnology research, quality control, or pharmaceutical development to apply scientific knowledge and grow within a research-driven organization.

EDUCATION

Master of Science (M.Sc.) — Biotechnology *Oct 2021 – Aug 2023*

The Assam Kaziranga University, Jorhat, Assam

Bachelor of Science (B.Sc.) — Biochemistry *Aug 2018 – Oct 2021*

Jagannath Barooah College, Jorhat, Assam

RESEARCH EXPERIENCE

M.Sc. Dissertation — Isolation of Endophytes from Mango Ginger (*Curcuma amada*) *June 2023*

The Assam Kaziranga University, Jorhat, Assam

- Conducted independent analytical research on the isolation and characterization of endophytic microorganisms from *Curcuma amada* (mango ginger) as partial fulfillment of M.Sc. in Biotechnology.
- Performed aseptic laboratory techniques including surface sterilization, tissue culture, microbial isolation, and subculturing on selective media.
- Documented and analyzed morphological characteristics of isolated endophytes for identification and classification.
- Compiled findings into a research dissertation submitted to the Department of Biotechnology, School of Basic Sciences.

TECHNICAL SKILLS

Laboratory Techniques: Microbial Culture & Isolation, Aseptic Technique, Surface Sterilization, DNA/RNA Extraction, Gel Electrophoresis, PCR, Spectrophotometry, Media Preparation

Computational & Bioinformatics: Molecular Docking, DNA Sequence Analysis, Gene Expression Profiling, Allele Frequency Calculation, Population Genetics Modeling, Python for Bioinformatics, Deep Learning (PyTorch)

Software & Tools: Microsoft Excel, Power BI, Python, PyTorch, OpenCV, BLAST, NCBI Databases

Core Knowledge: Molecular Biology, Microbiology, Biochemistry, Toxicology, Pharmaceutical Biotechnology, Stem Cell Biology, Immunology, Genetics

PROJECTS

PyTorch Melanoma Image Classifier *Python, PyTorch, OpenCV*

github.com/FarhanAlam2001/PyTorch_Melanoma_Image_Classifier

- Built a from-scratch convolutional neural network in PyTorch for benign vs. melanoma dermoscopic skin lesion classification, implementing the full pipeline from raw images to trained model and inference.
- Designed a 3-layer CNN architecture (Conv2D + ReLU + MaxPool) with a custom preprocessing pipeline (grayscale conversion, 50×50 resizing, normalization) and one-hot label encoding.
- Developed separate scripts for preprocessing, training (Adam optimizer), evaluation, and single-image inference with confidence scoring, alongside a full README documenting architecture, known limitations, and an explicit non-clinical-use disclaimer.

Cheminformatics Exercises *Python, SwissADME, ProTox-3.0, SwissDock, AutoDock Vina*

github.com/FarhanAlam2001/Cheminformatics_Projects

- Built a series of cheminformatics exercises covering ADMET property prediction, toxicity profiling, and protein–ligand docking for drug discovery workflows.
- Performed ADMET/toxicity analysis with SwissADME and ProTox-3.0 on a drug-like molecule (C₁₅H₂₃N₃O₂S), confirming Lipinski rule compliance with 0 violations.
- Conducted exhaustive molecular docking studies (SwissDock/AutoDock Vina) of testosterone against the 1B7T protein and chloroquine against the 3V6P protein, producing docked complex structures and full docking reports.

Python for Genetics Projects *Python (Standard Library)*

github.com/FarhanAlam2001/Python-For-Genetics-Projects

- Developed a collection of Python scripts for computational genetics, including DNA sequence analysis, transcription/translation simulation, and ORF finding.
- Implemented population genetics tools covering allele inheritance, allele frequency calculation, Hardy–Weinberg equilibrium, and recombination frequency estimation.
- Built sequence-comparison utilities (Hamming distance, molecule classification) and FASTA file parsing with algorithm performance benchmarking.

DNA Toolkit *Python*

github.com/FarhanAlam2001/DNA_Toolkit

- Created a beginner-friendly DNA sequence analysis toolkit supporting sequence validation, transcription (DNA → RNA), reverse complement generation, and protein translation via codon table.
- Added GC content calculation, codon usage frequency analysis, and random DNA sequence generation for testing and simulation.

Genome Toolkit *Python*

github.com/FarhanAlam2001/Genome-Toolkit

- Built a modular genome analysis toolkit providing nucleotide counting, Open Reading Frame (ORF) finding, motif/substring search, and k-mer frequency analysis.

Rosalind Solutions *Python*

github.com/FarhanAlam2001/Farhans_Rosalind_Solutions_Projects

- Solved a range of bioinformatics problems from Rosalind.info focused on DNA, RNA, and protein string manipulation, core bioinformatics algorithms, and file-based I/O handling.

Patient FASTQ QC Pipeline *Bash, FastQC, grep/awk*

github.com/FarhanAlam2001/patient-fastq-qc-pipeline

- Built a Bash automation pipeline that runs FastQC quality control on a patient's sequencing data and compiles a clean summary report from a single command.
- Automated FASTQ discovery (.fastq/.fq/.gz) per patient ID, parsed FastQC output with grep/awk to isolate FAIL-level metrics (per-sequence quality, GC content, overrepresented sequences), and generated timestamped, re-run-safe logs and reports.
- Implemented secure report handling with owner-only permissions (chmod 600) and strict error handling (set -euo pipefail) with defined exit codes for missing arguments, data, or dependencies.

INTERNSHIP EXPERIENCE

Toxicology Laboratory Associate (Online Internship) *Apr 1–14, 2024 | Grade: A (84.45%)*

BioLim Centre for Science & Technology, Chennai

- Completed a 14-day certified internship covering toxicological assessment methods, hazard evaluation, and laboratory safety protocols.
- Performed online tasks, implementation exercises, and viva assessments; achieved Grade A with distinction.

Pharmaceutical Biotechnologist (Online Internship) *Dec 23, 2023 – Jan 12, 2024 | Grade: B (76%)*

BioLim Centre for Science & Technology, Chennai

- Completed a 21-day certified internship focusing on pharmaceutical biotechnology concepts, drug development processes, and biomanufacturing principles.
- Engaged in structured tasks covering formulation, quality control, and regulatory aspects of pharmaceutical biotechnology.

CERTIFICATIONS

- Molecular Docking Certification Course — BioGrademy (BioNovura Labs LLP), Jan 2025
- Stem Cell Research Certification Course — BioGrademy (BioNovura Labs LLP), Dec 2024
- Internship on Toxicology Laboratory Associate Role — BioLim Centre for Science & Technology (BCIL Member), Apr 2024
- Internship on Pharmaceutical Biotechnologist Role — BioLim Centre for Science & Technology (BCIL Member), Jan 2024
- Introduction to Python for Genetics
- AI Dashboards Using Microsoft Power BI — Workshop

SOFT SKILLS

Communication | Teamwork | Problem Solving | Self-Directed Learning | Attention to Detail | Analytical Thinking

LANGUAGES

English (Professional) | Hindi (Native) | Assamese (Native)